

OSSP (Operating Systems And Systems Programming) - 25CS2104E

1. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
abhinav@Abhinav:~$ mkdir PROCESSID
abhinav@Abhinav:~$ nano PROCESSID.c
abhinav@Abhinav:~$ nano PROCESSID.c
abhinav@Abhinav:~$ gcc PROCESSID.c -o program
abhinav@Abhinav:~$ ./program
```

My Process ID (PID) = 654

OUTPUTS: abhinav@Abhinav:~\$

```
abhinav@Abhinav:~$ nano CHILDPROCESS.c
abhinav@Abhinav:~$ gcc CHILDPROCESS.c -o program
abhinav@Abhinav:~$ ./program
```

Parent Process

PID = 919

Child Process

PID = 920

```
PID = 920
abhinav@Abhinav:~$ mkdir PARENTPID
abhinav@Abhinav:~$ nano PARENTPID.c
abhinav@Abhinav:~$ gcc PARENTPID.c -o program
abhinav@Abhinav:~$ ./program
```

Child Process Executing...

Child Finished

Parent Process Continues

abhinav@Abhinav:~\$

Parent Process Continues

```
abhinav@Abhinav:~$ gcc MULTIPLECHILD.c -o program
abhinav@Abhinav:~$ ./program
```

Child 1 PID = 1081

Child 2 PID = 1082

Child 3 PID = 1083

abhinav@Abhinav:~\$